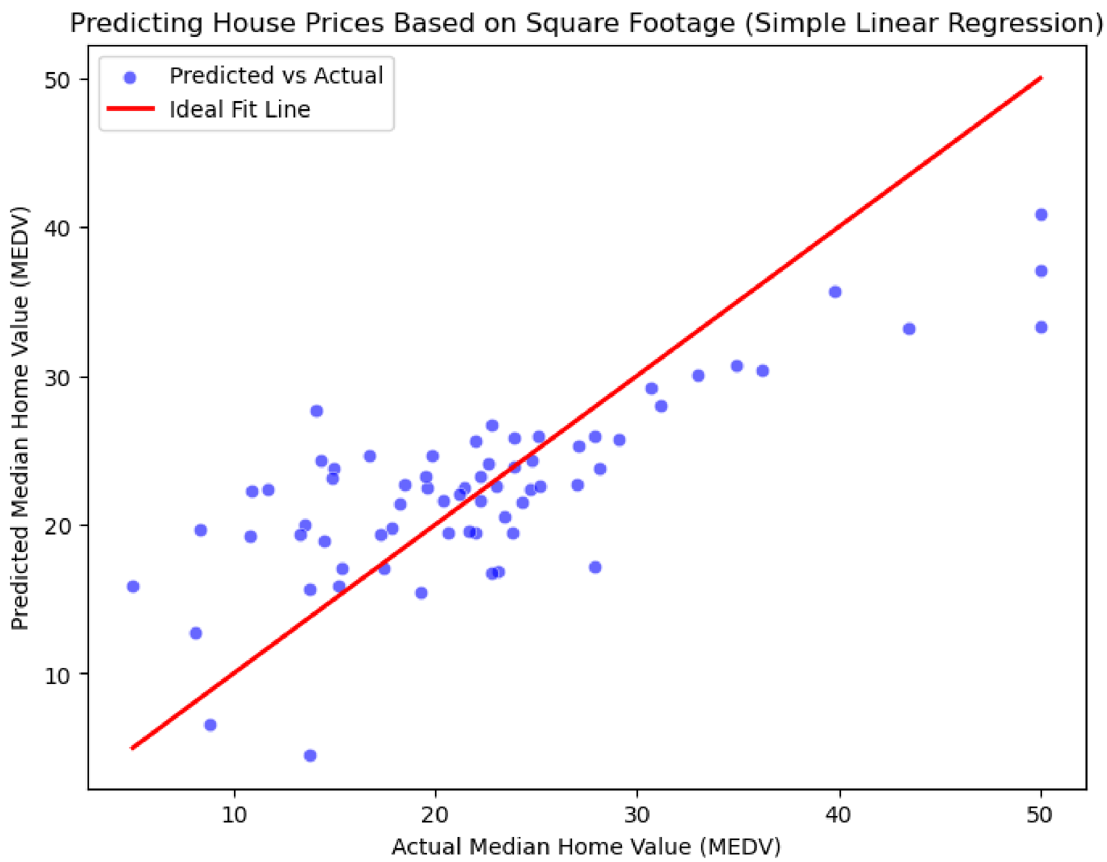
simplelinearregression

November 2, 2025

[12]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# Load dataset
train = pd.read_csv(r"D:\Projects and␣
↪All\gitupload\upload-folders\DataScienceProject2\DataScienceProject\Machine-and-Deep-Learni
↪csv")

# Select relevant columns
X = train[['rm']]   # Predictor
y = train['medv']   # Target

# Split into train/validation sets
X_train, X_val, y_train, y_val = train_test_split(X, y, test_size=0.2,␣
↪random_state=42)

# Train model
model = LinearRegression()
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_val)

# Evaluate
print("R² Score:", r2_score(y_val, y_pred))
print("MSE:", mean_squared_error(y_val, y_pred))

# Plot
plt.figure(figsize=(8,6))
sns.scatterplot(x=y_val, y=y_pred, color='blue', alpha=0.6, label='Predicted vs␣
↪Actual')
plt.plot([y_val.min(), y_val.max()], [y_val.min(), y_val.max()], color='red',␣
↪lw=2, label='Ideal Fit Line')
```

```
plt.xlabel("Actual Median Home Value (MEDV)")
plt.ylabel("Predicted Median Home Value (MEDV)")
plt.title("Predicting House Prices Based on Square Footage (Simple Linear␣
  ↪Regression)")
plt.legend()
plt.show()

# Model coefficients
print(f"Intercept: {model.intercept_:.2f}")
print(f"Coefficient (effect of RM): {model.coef_[0]:.2f}")
```

```
R² Score: 0.5959747117709422
MSE: 36.361622515889756
```




```
Intercept: -30.96
Coefficient (effect of RM): 8.58
```

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